Transformer

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **High** 1 labs

Covers these lab items: Small size/demonstration units of transformer

What it is used for

A transformer transfers electrical energy between circuits through electromagnetic induction and may change AC voltage levels. Use demonstration units only within their rated voltage, current, and power limits.

How to use

1. Inspect the transformer, terminals, insulation, core, and connecting leads for damage.
2. Read the nameplate to identify input voltage, output voltage, frequency, and rating.
3. Ensure the power supply is switched off before making connections.
4. Connect the primary and secondary circuits according to the demonstration diagram.
5. Check that the load and measuring instruments are suitable for the transformer rating.
6. Switch on the supply and measure the output using correctly rated instruments.
7. Record the readings and switch off the supply before changing connections.
8. Disconnect leads safely and store the transformer after the unit cools if necessary.

⚠ Safety

- Never touch exposed terminals while the transformer is energised.
- Do not exceed rated voltage, current, or power limits.
- Use qualified supervision and appropriate electrical protection during demonstrations.
- Disconnect power immediately if smoke, unusual noise, overheating, or burning smell occurs.

Care & storage

- Keep the transformer dry, clean, and free from obstructed ventilation.
- Inspect leads and insulation regularly and report damaged equipment before use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.